

## Highlights

**A cortical-proxy observable for emotionally manipulative media: an instrument, a 400-item corpus, and the field bound it yields**

Brian Mwai

- A released cortical encoder turns text into a signed affective-deliberative index
- Four length-matched corpora separate at eta-squared 0.107 and replicate in a rerun
- The separation is between source datasets; a TF-IDF baseline far outperforms it
- High-outrage text raises both networks equally and does not separate from neutral
- The measured spread 0.124 requires a coupling of at least 4.29 at beta  $J = 2$

# A cortical-proxy observable for emotionally manipulative media: an instrument, a 400-item corpus, and the field bound it yields

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## Abstract

Mean-field opinion models represent media as an external field acting on coupled binary opinions, and that field is almost never measured. A companion paper shows that a media-driven transition requires a coupling  $\alpha \geq h_c(\beta J)/\Delta X$ , where  $\Delta X$  is the attainable range of a content observable. Here we measure one. Text is converted to speech, aligned at word level and passed to a released encoder that predicts average-subject fMRI responses on 20,484 cortical vertices; the observable is the mean predicted activation over an affective-salience parcel set (1,030 vertices) minus that over a deliberative-control set (851 vertices). No person was scanned. On 400 length-matched items in four categories, each drawn from a single source dataset, the categories separate at  $\eta^2 = 0.1068$  ( $F = 15.779$ ,  $p = 1.0 \times 10^{-9}$ ) against a smallest detectable  $\eta^2 = 0.0268$ . The separation is carried by fear-activating items (Cohen's  $d = 0.94$  against neutral); high-outrage items do not separate ( $d = 0.03$ ) because they raise both networks almost equally. Because category and source coincide, the separation is between corpora and cannot be attributed to framing. Against the source-derived manipulative label the observable reaches AUC = 0.627 while a TF-IDF baseline reaches 0.976, so it is offered as a field observable, not a detector. A second run gives ICC(2,1) = 0.8725 and replicates the separation ( $\eta^2 = 0.0888$ ), but 51 of 400 items change sign, as the session noise predicts, so no per-item statement is supported. The measured spread  $\Delta X = 0.1241$  implies  $\alpha \geq 4.29$  at

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$\beta J = 2$ .

*Keywords:* sociophysics, opinion dynamics, external field, mean-field Ising model, brain encoding model, test-retest reliability

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## 1. Introduction

Statistical-physics treatments of opinion formation routinely include a term representing media. In the mean-field Ising formulation and its relatives [1, 2], that term is an external field  $h$  acting on every agent, and the modelling result is that beyond a critical magnitude  $h_c(\beta J)$  the population is driven from one consensus to another. The literature on this mechanism is large [3, 4, 5, 6] and the qualitative conclusion is robust within the models.

The field itself is almost never measured. It is assigned, swept, or fitted to an outcome that the model was built to reproduce, and a recent review of Ising-inspired sociophysics lists calibration against data among the field’s open problems [7]. A companion paper [8] makes the consequence precise: writing  $h = \alpha X$  for a content observable  $X$  with attainable range  $\Delta X$ , no media-driven transition across the spinodal of the all-to-all mean-field model is possible unless

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}, \quad (1)$$

and this can be evaluated before any data are collected. What it cannot supply is  $\Delta X$ , which is an empirical property of a real observable on real content.

This paper supplies one. We build an instrument that computes a candidate observable from content, apply it to a corpus assembled for the purpose, and report the spread it attains together with what the measurement does and does not support. Three commitments run through the work and are stated here so the reader can hold us to them: the physics stays in front and the encoder is treated as a thermometer rather than a subject; every number is produced by a released script from released data; and the design’s confound is stated before each result it touches, not deferred.

## 2. Methods

### 2.1. Instrument

Each item is converted to speech, transcribed with word-level timings, embedded as text and as audio, and passed to TRIBE v2 [9], a released mul-

timodal encoder, developed from the TRIBE architecture [10], that predicts fMRI responses over 20,484 vertices of the fsaverage5 cortical surface [11]. Speech is synthesised with gTTS [12]; when that service is unreachable the encoder’s own loader falls back to the MMS-TTS English model [13], so two runs of the pipeline need not use the same synthesiser. Word timings come from WhisperX [14], which aligns the output of Whisper [15]. The encoder embeds text with Llama 3.2-3B [16] and audio with Wav2Vec-BERT 2.0 [17]. The released checkpoint is loaded in the average-subject configuration that its loader sets, so every prediction is for a group-average subject rather than for any individual. The prediction, an array of shape  $(T, 20,484)$  over the item’s duration, is mean-pooled over time to one value per vertex. We use the word *scan* for one pass of the encoder over an item or over the corpus; no person was scanned at any point.

Two parcel sets are taken from the HCP multimodal parcellation (HCP-MMP1.0) [18], bilaterally, and resolved once into fsaverage5 vertex index sets using the encoder’s own atlas utility. The affective-salience set is OFC, pOFC, p24, a24, TGd, TE1a, TE1p, IFSa, IFSp and AAIC (1,030 vertices); the deliberative-control set is 46, a9-46v, p9-46v, 11l, 13l, d32, p32, a10p, p10p and 10pp (851 vertices). The sets do not overlap (Fig. 1). Their assignment to the two labels is the study’s specification, made before the data existed, and is not an independently validated functional division.

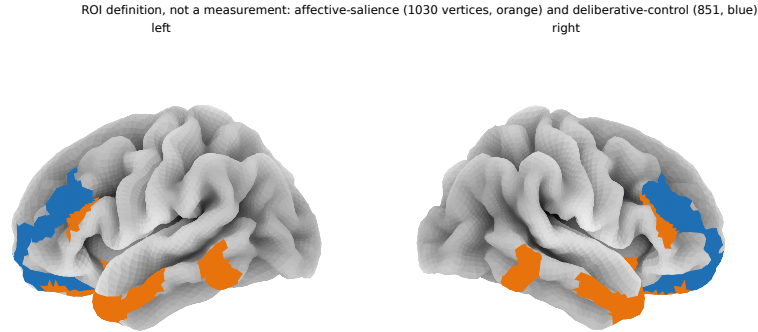


Figure 1: The two parcel sets the observable is reduced over, on the cortical surface. The affective-salience set covers 1030 vertices and the deliberative-control set 851. Both are resolved once into vertex index sets, so every item is averaged over identical regions. The figure is a definition and carries no data.

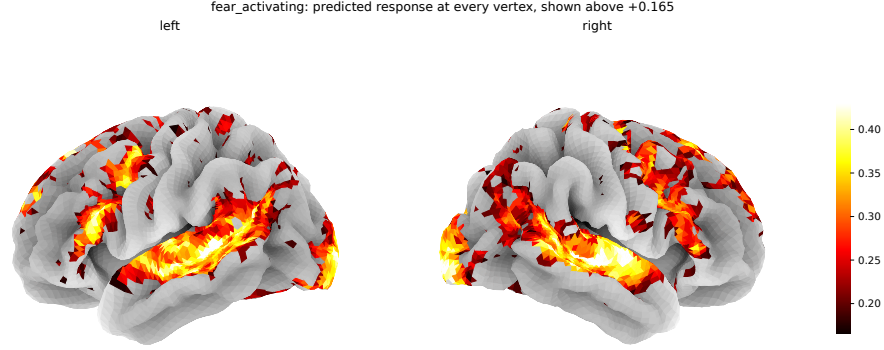


Figure 2: Example of the encoder’s output: the time-averaged predicted response for one fear-activating item at every one of the 20,484 cortical vertices, on the pial surface. Vertices below the map’s 70th percentile are left unpainted so the anatomy remains readable, and the threshold is stated in the figure. The map is shown to illustrate the instrument, not as a statement about this item (Section 3.4).

#### 2.1.1. The observable

Let  $A_{\text{aff}}$  and  $A_{\text{del}}$  be the mean time-averaged predicted activation over the affective-salience and deliberative-control vertex sets. The original specification of this project used a ratio,

$$\text{NAA}_{\text{ratio}} = \frac{A_{\text{aff}}}{A_{\text{del}} + \delta}, \quad \delta = 10^{-3}, \quad (2)$$

which is a magnitude asymmetry only when both means sit at or above baseline. The encoder’s outputs are unconstrained in sign, and a negative denominator makes the ratio change sign for reasons that have nothing to do with the content. Across the corpus reported below the ratio is undefined for 69 of 400 items, 17.25%. This is not an edge case to be dropped, so the analysis is carried by the signed difference

$$X \equiv \text{NAA}_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}, \quad (3)$$

which is defined everywhere. Positive means affective activation is predicted to lead, negative means deliberative activation leads. The reformulation changes what the index means: it reports which set leads, and by how much in the encoder’s units, not by what proportion.

The observable is a *cortical proxy*. The released checkpoint predicts activity on the cortical surface only, so no subcortical structure is measured and none is named. Every value is a prediction produced by an encoder from content, and the phrase “predicted activation” is used throughout in place of any claim about measured brain activity.

## 2.2. Corpus

The corpus is 400 English text items in four categories of 100 (Table 1). Each item is a passage of whole sentences between 150 and 200 words, so that category is not confounded with the length of the encoder’s input. Mean word counts are 167.4, 163.5, 163.7 and 162.2 for fear-activating, high-outrage, neutral and reward-hook items respectively, with standard deviations between 10.2 and 12.8.

| Category              | Source                           | n       |
|-----------------------|----------------------------------|---------|
| Fear activating       | ISOT fake news [19]              | 100     |
| High outrage          | SemEval-2019 Task 4 [20]         | 100     |
| Reward hook           | Webis-Clickbait-17 [21]          | 100     |
| Neutral informational | PubMed [22], ISOT true news [19] | 50 + 50 |

Table 1: Corpus composition. Each category is drawn from exactly one source dataset (two for neutral), which is the study’s dominant limitation and is stated before every result it affects.

The category names state the construct each source was chosen to represent; they are not per-item annotations. Fear-activating items are articles from the ISOT fake-news collection [19] whose title or body contains at least one of 24 threat keywords (for example “outbreak”, “attack”, “toxic”). High-outrage items are articles labelled hyperpartisan in the SemEval-2019 Task 4 by-publisher collection [20], whose labels are inherited from the publisher rather than assigned per article. Reward-hook items are the linked-article bodies of Webis-Clickbait-17 posts judged clickbait [21]; the teaser itself is about ten words and too short to be length-matched. Neutral items are a seeded random sample of English journal-article abstracts from PubMed published 2019–2021 [22], retrieved without subject terms, and Reuters wire articles from the ISOT true-news collection. The ISOT true articles open with a Reuters dateline that the fake articles lack, and a classifier can separate the two on that string alone; datelines and residual mentions of the agency were

therefore removed before passages were built. The binary **manipulative** label used in Section 3.3 is derived from the category (1 for the three non-neutral categories, 0 for neutral), not annotated per item. The working hypothesis, fixed before scanning, was that the three non-neutral categories would sit above neutral content on  $X$ , high outrage most of all.

**The corpus is not source-balanced, and this is the central caveat of the paper.** Category and provenance coincide by construction, so any separation the instrument reports is a separation between corpora and cannot be attributed to framing. Section 3 states this alongside the result rather than after it, and Section 4.1 identifies which conclusions survive it.

### 2.3. Execution and the second run

The corpus was shuffled with a fixed seed before scanning, so items were scanned interleaved rather than one category at a time, and each result was appended to the output as it completed. The first run (A) took four GPU sessions on a Tesla P100. Every row was checked against the source corpus on text, category and identifier before analysis. A second run (B) rescanned all 400 items with identical text, identical code and identical parcel sets. Run A is the primary measurement; run B is used only for reliability and replication. Neither run is corrected toward the other and the two are not averaged, since an average of two runs would carry a precision that neither has.

### 2.4. Statistical analysis

Separation across the four categories is tested by one-way analysis of variance on  $X$ , reporting  $F$ , its  $p$ -value and  $\eta^2 = SS_{\text{between}}/SS_{\text{total}}$ . Each non-neutral category is compared with neutral content by Welch’s unequal-variance  $t$ -test [23], with effect size Cohen’s  $d = (\bar{X}_1 - \bar{X}_0)/s_p$ , where  $s_p$  is the pooled standard deviation [24]; with 100 items per group,  $s_p^2$  is the mean of the two sample variances. The same contrasts are repeated on  $A_{\text{aff}}$  and  $A_{\text{del}}$  separately. Every  $p$ -value is the probability, computed under the null hypothesis of equal means (or no discrimination), of a test statistic at least as extreme as the one observed; it is not the probability that the null hypothesis is true.

Discrimination against the **manipulative** label is summarised by the area under the ROC curve,  $\text{AUC} = P(X_+ > X_-) + \frac{1}{2}P(X_+ = X_-)$  for a randomly chosen labelled-positive item  $X_+$  and labelled-negative item  $X_-$ , which equals the Mann–Whitney  $U$  statistic divided by  $n_+n_-$  [25, 26]. No parameter is

fitted for the observable. Two baselines are scored on the same rows and label: the VADER compound sentiment score [27], as implemented in NLTK [28], used directly; and TF-IDF features over unigrams and bigrams (sublinear term frequency, minimum document frequency 2) with logistic regression, implemented in scikit-learn [29] and scored out of fold under five-fold stratified cross-validation. Confidence intervals on AUC are percentile intervals from 2,000 bootstrap resamples of items [30]. A source-identity probe fits the same TF-IDF pipeline to predict the source dataset, scored out of fold.

Test–retest agreement between runs A and B is reported as Pearson  $r$  and as the intraclass correlation ICC(2,1): two-way random effects, absolute agreement, single measurement [31, 32]. ICC is reported because a run that shifted every score by a constant would still correlate perfectly. The number of items whose sign of  $X$  differs between runs is compared with the number expected under additive, independent per-run error, estimated from the paired differences (Section 3.4).

Sensitivity was computed from the design alone. For the ANOVA, the smallest  $\eta^2$  detectable at  $\alpha = 0.05$  with power 0.80 follows from the noncentral  $F$  distribution with noncentrality  $\lambda = N\eta^2/(1 - \eta^2)$  [24]; for the AUC, from a seeded simulation of a two-sided Mann–Whitney test under a binormal alternative with 300 positives and 100 negatives. Computations used SciPy [33].

### 2.5. Field model

The field enters the all-to-all mean-field model through the self-consistency condition  $m = \tanh(\beta J m + h)$ , with  $h$  in units of  $k_B T$ . For  $\beta J > 1$  the spinodal field at which the metastable branch disappears is

$$h_c(\beta J) = \sqrt{\beta J(\beta J - 1)} - \operatorname{artanh} \sqrt{1 - 1/\beta J}, \quad (4)$$

which reproduces the values located numerically in the companion paper [8] to the precision quoted here. The spread entering Eq. (1) is the sample range  $\Delta X = \max X - \min X$  over the corpus in run A.

## 3. Results

### 3.1. Separation across categories

The four categories differ at

$$F(3, 396) = 15.779, \quad p = 1.035 \times 10^{-9}, \quad \eta^2 = 0.1068, \quad (5)$$

on  $n = 400$ . The design can detect  $\eta^2 = 0.0268$  at  $\alpha = 0.05$  and power 0.80, so the observed effect is roughly four times the smallest the study could have found.

*Read this with Table 1 in view.* The categories are corpora. The measurement establishes that the instrument assigns systematically different values to four groups of documents; it does not establish that framing is what it responds to.

| Category vs neutral | t      | p                       | Cohen's d |
|---------------------|--------|-------------------------|-----------|
| Fear activating     | +6.639 | $3.283 \times 10^{-10}$ | +0.939    |
| Reward hook         | +2.254 | $2.539 \times 10^{-2}$  | +0.319    |
| High outrage        | +0.212 | $8.320 \times 10^{-1}$  | +0.030    |

Table 2: Welch comparisons of the observable  $X$  against neutral informational content,  $n = 100$  per group. The separation is carried by fear-activating items. High-outrage items do not separate, and that is the category the study’s hypothesis expected to separate most.

Two features of Table 2 deserve emphasis. First, the effect is largely one category: fear-activating items carry most of it. Second, high outrage does not separate on the observable despite being drawn from its own distinct source. Section 3.2 shows why, and the answer is not that outrage content leaves the encoder unmoved.

Every category mean is negative, so deliberative activation is predicted to lead on average in every category (Fig. 3). The instrument finds no affective dominance at category level anywhere in this corpus.

### 3.2. The two components, taken separately

The observable is a difference, and a difference cannot tell a large symmetric response from no response. Both components are recorded per item, so each can be compared against neutral content on its own (Table 3, Fig. 4).

High outrage raises  $A_{\text{aff}}$  by  $d = +0.507$  and  $A_{\text{del}}$  by  $d = +0.491$ , both at  $p < 10^{-3}$ , and the difference between them by  $d = +0.030$ . The category that fails to separate on the observable is the one that moves both networks together. Fear-activating content raises  $A_{\text{aff}}$  by  $d = +0.613$  and leaves  $A_{\text{del}}$  flat, so its whole effect on the observable is one-sided. A one-way ANOVA on each component alone gives  $F = 7.451$  for  $A_{\text{aff}}$  and  $F = 6.399$  for  $A_{\text{del}}$ , both  $p < 10^{-3}$ .

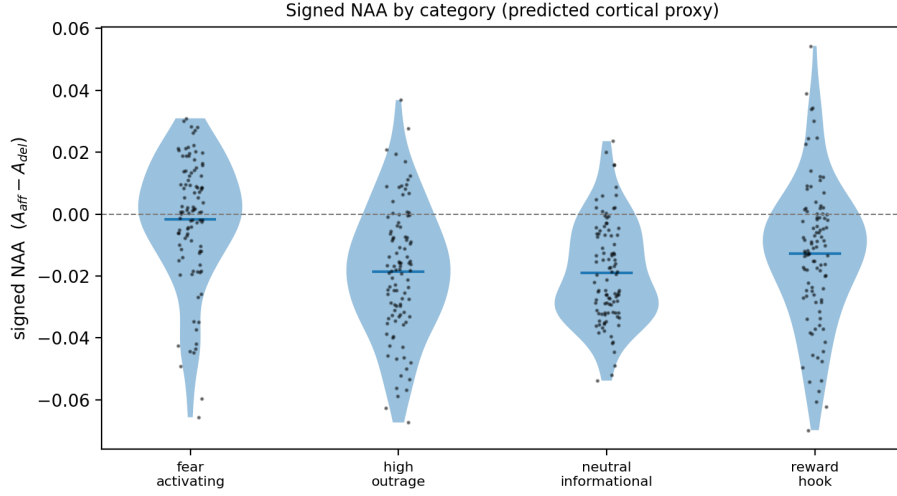


Figure 3: Distribution of the signed observable  $X$  within each category, all 400 items of run A; bars mark category means. Every category mean is negative, and the separation is a shift between overlapping distributions rather than a clean split.

| Category vs neutral | Component        | t      | p                      | Cohen's d |
|---------------------|------------------|--------|------------------------|-----------|
| Fear activating     | $A_{\text{aff}}$ | +4.333 | $2.365 \times 10^{-5}$ | +0.613    |
| Fear activating     | $A_{\text{del}}$ | -0.696 | $4.875 \times 10^{-1}$ | -0.098    |
| High outrage        | $A_{\text{aff}}$ | +3.584 | $4.263 \times 10^{-4}$ | +0.507    |
| High outrage        | $A_{\text{del}}$ | +3.468 | $6.521 \times 10^{-4}$ | +0.491    |
| Reward hook         | $A_{\text{aff}}$ | +1.842 | $6.691 \times 10^{-2}$ | +0.261    |
| Reward hook         | $A_{\text{del}}$ | -0.063 | $9.501 \times 10^{-1}$ | -0.009    |

Table 3: Welch comparisons of each component against neutral informational content. High outrage moves both components by nearly the same amount; fear-activating content moves one.

This weakens an argument we would otherwise make. If outrage had moved neither component, its failure to separate would count against the instrument simply detecting provenance. It moves both, so every category drawn from a distinct source moves away from the neutral baseline on at least one component, which is what source detection predicts. The components do not resolve the confound in either direction.

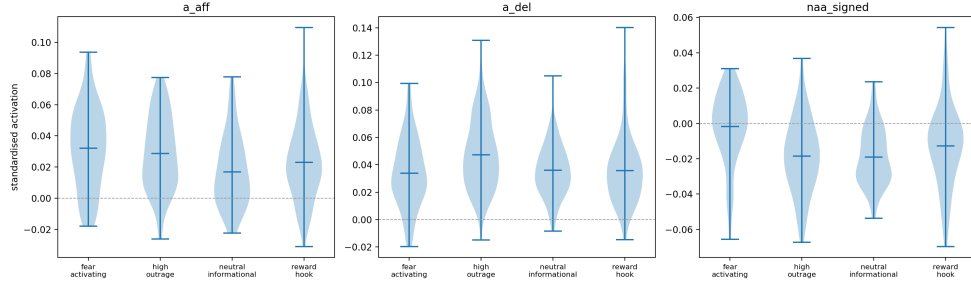


Figure 4:  $A_{\text{aff}}$ ,  $A_{\text{del}}$  and their signed difference by category over all 400 items of run A. High outrage is elevated on both components and flat on the difference.

### 3.3. Discrimination against the label, and baselines

Against the source-derived **manipulative** label (300 positive, 100 negative), the observable discriminates at  $\text{AUC} = 0.6274$ , above the smallest detectable 0.5916. Because the label is the category, this is a second view of the separation above rather than evidence about manipulation in individual items. AUC is the headline because it is threshold-free. An  $F_1$  of 0.6680 is obtainable, but its threshold was chosen on the same data it is evaluated on, and it is labelled as fitted in sample wherever it appears.

**The lexical baseline wins by a wide margin** (Table 4). Sentiment does not clear chance. The observable is not a better detector of manipulative content than counting words, and no such claim is made. Leave-one-source-out, the natural correction for the confound, is unavailable because every source contributes exactly one class, so a held-out source leaves a single-class test fold. The ISOT-only block is the substitute, and TF-IDF holds at 0.9724 there. Both TF-IDF figures sit above 0.95, where near-ceiling lexical performance on a small corpus points to residual stylistic markers, and a probe on the same features recovers the source dataset for 66.25% of items against a majority-class rate of 25%. Much of what the lexical

| Model             | Split             | n   | AUC    | 95% CI           |
|-------------------|-------------------|-----|--------|------------------|
| Observable $X$    | nothing fitted    | 400 | 0.6274 | [0.5702, 0.6867] |
| VADER compound    | nothing fitted    | 400 | 0.5392 | [0.4757, 0.6035] |
| TF-IDF + logistic | 5-fold stratified | 400 | 0.9758 | [0.9601, 0.9879] |
| Observable $X$    | ISOT only         | 150 | 0.7290 | [0.6431, 0.8132] |
| VADER compound    | ISOT only         | 150 | 0.4019 | [0.3098, 0.4943] |
| TF-IDF + logistic | ISOT only, 5-fold | 150 | 0.9724 | [0.9483, 0.9901] |

Table 4: Baselines against the same label. Intervals are 2,000-draw bootstrap percentiles. The lower block keeps only the 150 items from the ISOT collection, the one origin corpus contributing both classes.

model separates is provenance. The observable is carried forward as the field quantity Eq. (1) consumes, not as a competing classifier.

### 3.4. Run-to-run reliability

Without a second run every effect above would rest on a single pass with no statement of how much of an item’s score is the item and how much is the run. Table 5 gives the agreement between runs A and B.

|                                  | $r$    | ICC(2,1) | sd( $\Delta$ ) | sd( $\Delta$ )/sd <sub>A</sub> |
|----------------------------------|--------|----------|----------------|--------------------------------|
| $X = \text{NAA}_{\text{signed}}$ | 0.8763 | 0.8725   | 0.01029        | 0.484                          |
| $A_{\text{aff}}$                 | 0.8759 | 0.8713   | 0.01213        | 0.489                          |
| $A_{\text{del}}$                 | 0.8576 | 0.8550   | 0.01271        | 0.521                          |

Table 5: Agreement between two runs over all 400 items.  $\Delta$  is the paired difference B–A and sd<sub>A</sub> the between-item standard deviation in run A. ICC is reported beside  $r$  because a run that shifted every score by a constant would still correlate perfectly.

Three consequences follow, and they cut in different directions.

**The separation replicates.** On run B the four-category ANOVA gives  $\eta^2 = 0.0888$ ,  $F = 12.861$ ,  $p = 4.918 \times 10^{-8}$ , against  $\eta^2 = 0.1068$ ,  $F = 15.779$  for run A over the same items. Both clear the detectable floor. The difference in  $\eta^2$  between runs is 0.018, with a paired bootstrap 95% interval of  $[-0.014, 0.052]$  (5,000 resamples of items within category), so the two runs are consistent with one underlying effect measured twice with error.

**Per-item direction is not reliable.** Of 400 paired items, 51, or 12.75%, reverse sign between runs. A statement that a particular item is affective-leading is therefore not supportable at this precision, and no per-item verdict appears anywhere in this paper or in the public interface described under Data availability. Group means over 100 items average this scatter down by a factor of ten, which is why the category-level result survives.

**The reversal count is the measured noise, not an extra defect.** The paired differences give a per-run error  $\sigma = \text{sd}(\Delta)/\sqrt{2} = 0.00728$ , against a between-item standard deviation of 0.01913 once that error is removed. Under additive error at that  $\sigma$ , with each item’s value shrunk toward the grand mean so that no item’s own noise predicts its own reversal, 20,000 simulations give 55.3 expected reversals, 95% interval [44, 67]. The observed 51 sits inside it. Reversing items sit near zero, mean  $|X| = 0.00320$  against 0.02159 for stable items (Mann–Whitney  $p = 1.4 \times 10^{-25}$ ), and spread evenly across categories ( $\chi^2 = 2.764$ , 3 df,  $p = 0.429$ ). Per-item direction is therefore out of reach at this  $\sigma$ , rather than awaiting a fix. Three sources could contribute to  $\sigma$ : speech synthesis, which is redone on every run and may fall back to a different synthesiser; forced alignment; and GPU nondeterminism. Separating them requires a controlled rerun.

**Scan order does not carry the separation.** Because run A took more than one GPU session, position in the scan is a candidate explanation for the category effect. The interleaved order balances the four categories across thirds of the scan ( $\chi^2 = 2.773$ , 6 df,  $p = 0.837$ ). A drift across thirds is present,  $\eta^2 = 0.0167$  at  $p = 0.0353$ , but because it is nearly orthogonal to category it adds scatter rather than bias: removing scan-third means before the category ANOVA moves the effect only from  $\eta^2 = 0.1068$  to 0.1048,  $p = 1.588 \times 10^{-9}$ . Had the corpus been scanned one category at a time, this drift would have been indistinguishable from the finding.

### 3.5. *The field: calibration and bound*

With the observable in hand, the natural step is to estimate  $\alpha$  in  $h = \alpha X$  by fitting. We regressed the **manipulative** label on  $X$  (320 items for fitting, 80 held out) and converted the slope  $s$  with the linear-response relation  $m \approx h/(1 - \beta J)$ , valid for  $\beta J < 1$ , into  $\hat{\alpha} = s(1 - \beta J)$ . The bootstrap interval for  $\hat{\alpha}$  excludes zero. That is not a result, for three reasons. The outcome is a source-derived label, not a measured opinion. The fit performs worse than the sample mean on held-out data,  $R^2_{\text{held-out}} = -0.031$ , so the relationship does not generalise beyond the points used to estimate it. And  $\hat{\alpha}$  is proportional

to  $1 - \beta J$ , which nothing in this study measures, so any single value would be a statement about a chosen  $\beta J$  rather than about media.

The coupling is therefore *unidentified*, which is a weaker claim than either a detection or a null, and **no value of  $\hat{\alpha}$  is quoted in this paper**. Fig. 5 shows instead how the free energy would respond at each category’s measured mean for a range of assumed couplings.

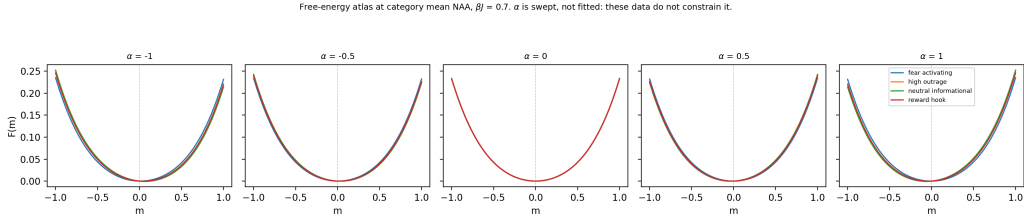


Figure 5: Landau free energy  $F(m)$  at  $\beta J = 0.7$ , evaluated at each category’s measured mean  $X$  for assumed couplings  $\alpha \in \{-1, -0.5, 0, 0.5, 1\}$ . At this coupling the model is below its critical point and each curve has a single minimum; the panels show how little the category means tilt it. The sweep is not a fit: no value of  $\alpha$  is estimated from the data.

The measurement does deliver the quantity Eq. (1) consumes. Across the corpus the observable attains

$$\Delta X = 0.1241 \quad (\min X = -0.0698, \max X = +0.0543), \quad (6)$$

and substituting into the companion paper’s inequality with Eq. (4) converts a general statement into a concrete requirement (Table 6).

| $\beta J$ | $h_c(\beta J)$ | $\alpha$ required |
|-----------|----------------|-------------------|
| 1.102     | 0.02101        | 0.1693            |
| 1.496     | 0.20541        | 1.6553            |
| 2.000     | 0.53284        | 4.2938            |

Table 6: The field bound evaluated at three reduced couplings (grid points of the companion paper’s solver nearest 1.1, 1.5 and 2), using the measured spread  $\Delta X = 0.1241$  of run A. Each row states the coupling any proposal of this mechanism would need in order for media to drive a transition through this observable.

This is a necessary condition and not a sufficient one. Clearing it does not establish that media moves populations; falling below it excludes the

mechanism within the model. Run B gives a smaller spread,  $\Delta X = 0.1176$ , and hence  $\alpha \geq 4.530$  at  $\beta J = 2$ . Because a sample range is inflated by measurement error, the noise-free spread is no larger than the measured one and the requirement it implies is no smaller, so the quoted values are the lenient end.

## 4. Discussion

### 4.1. What survives the confound

It is worth separating the claims this study can defend from those it cannot, because the two are easy to conflate.

*Not supported.* That the instrument detects manipulation, or does so better than a lexical model. That framing rather than provenance drives the separation. Any value for the coupling. Any statement about an individual item. Any claim about subcortical structures, or about measured rather than predicted activity.

*Supported.* That the observable attains a measured spread of about 0.12 on real content, which is what the bound consumes and which does not depend on why the spread arises. That the categories separate at an effect four times the detectable floor, and that the separation replicates in an independent run. That high-outrage content, the category the hypothesis favoured most, raises both networks together and so leaves the observable where neutral content puts it. And that the coupling is unidentified on this data, which constrains what a successor may assume.

The confound is removable by design rather than by analysis. Drawing all four categories from a single source, so that provenance is held constant while framing varies, or crossing sources with categories so that the two are statistically separable, would convert the present result into an interpretable one. Neither requires new instrumentation.

### 4.2. Limitations

1. Category is confounded with source dataset, and the `manipulative` label is derived from category. This is the dominant limitation, stated in Section 3 as well as here. Category membership is also partly defined by selection rules (threat keywords for fear-activating items, publisher-level labels for high-outrage items) rather than by per-item annotation.

2. Single-run scores are noisy:  $\text{ICC}(2, 1) = 0.8725$ , with run-to-run standard deviation 48.4% of the between-item standard deviation. Category-level conclusions survive this; per-item ones do not.
3. Predicted activation is not measured activation. Whether the released average-subject checkpoint tracks measured cortical responses to synthesised speech of written text is not tested here and is not established.
4. Cortical proxy only. The checkpoint cannot speak to subcortical structure.
5. Average-subject configuration, set by the loader; no individual differences are represented.
6. Out-of-distribution stimuli: the encoder was trained on naturalistic audiovisual material and is given synthesised speech from written text, with a synthesiser that may differ between runs.
7. The parcel sets are the study’s specification. The affective-salience set contains orbitofrontal parcels and the deliberative-control set lateral orbitofrontal ones, and other defensible choices would give a different observable.
8.  $\Delta X$  is the range of one run on this corpus. It depends on corpus size and composition, and differs between runs (0.1241 and 0.1176).
9. The corpus is English-language from five specific sources, none of them African. A search for openly licensed African news data carrying usable labels returned nothing suitable, so the spread is a property of American and European publishing and the bound inherits that limitation.
10. The field model is mean-field, single-population, static and binary.

## 5. Conclusion

Media enters mean-field opinion models as a field that is almost never measured. This paper measured a candidate: an observable computed from content through a released cortical encoder, applied to 400 items, with its spread, its reliability and its failure modes reported rather than assumed. The instrument separates four corpora at an effect it was powered to find, the separation replicates across runs, and it is confounded with provenance in a way no analysis of this corpus can undo. A lexical baseline separates the same corpora far better. The coupling the observable was built to calibrate is unidentified. What survives is the spread, and through the companion paper’s inequality a requirement,  $\alpha \geq 4.29$  at  $\beta J = 2$ , that any future proposal of this mechanism must meet within all-to-all mean-field theory.

## **CRedit authorship contribution statement**

**Brian Mwai:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

## **Declaration of competing interest**

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## **Funding**

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. Computation used free-tier GPU resources.

## **Ethics statement**

This study involved no human participants, no animal subjects and no personal data. Every activation value is a model prediction. The corpus consists of publicly released datasets and public PubMed abstracts, used under their respective terms. TRIBE v2 is released under the CC-BY-NC-4.0 licence and was used for non-commercial research only.

## **Data availability**

The pipeline, corpus builder, analysis and figure scripts, and the JSON and CSV files every number in this paper is read from are public at <https://github.com/brn-mwai/monarch>. The analyses are produced by `analyze_corpus.py` (separation, effect sizes, AUC), `outrage_split.py` (components), `baselines.py` (baselines and source probe), `test_retest.py` and `reversal_diagnosis.py` (reliability), `power_statement.py` (sensitivity), `calibrate_alpha.py` (calibration) and `field_bound.py` (bound), and `verify_numbers.py` checks every quoted value against those outputs. The full corpus, with each item’s text, its pre-scan labels and its measured values, is browsable at <https://monarch-4iy.pages.dev/corpus>; the interface reports no per-item verdict, for the reason given in Section 3.4. Figures of predicted responses are rendered from the saved per-vertex maps, each stating its threshold. The source datasets are available from their original providers [19, 20, 21, 22].

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI assistant (Claude, Anthropic) for editing text and for support in writing and checking analysis code. After using this tool, the author reviewed and edited all content and takes full responsibility for the content of the publication.

## Acknowledgements

The author thanks Dr. Songa Mutambi, Department of Natural Sciences, The Catholic University of Eastern Africa, who supervised the project of which this work forms part and suggested the signed-difference form of the observable used throughout this paper.

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